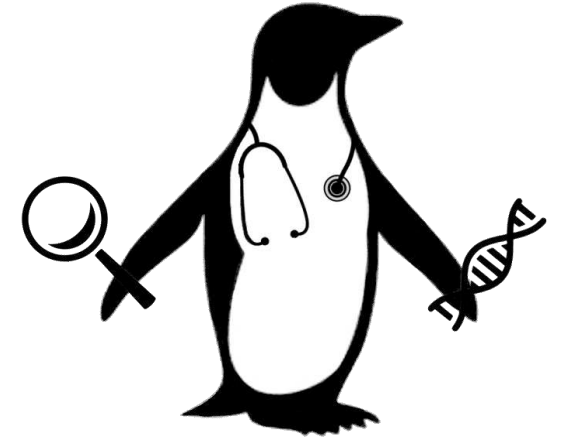




PENGQUIN:

PErsoNal Genome QUery IN healthcare and clinical practice

Elias Crum

European Semantic Web Conference
Ph.D. Symposium (Early Stage)
May 27th, 2024

Outline

Why?



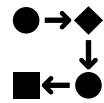
Genomic Data + Semantic Web?

Goals



Reshape how genomic data is stored

How?

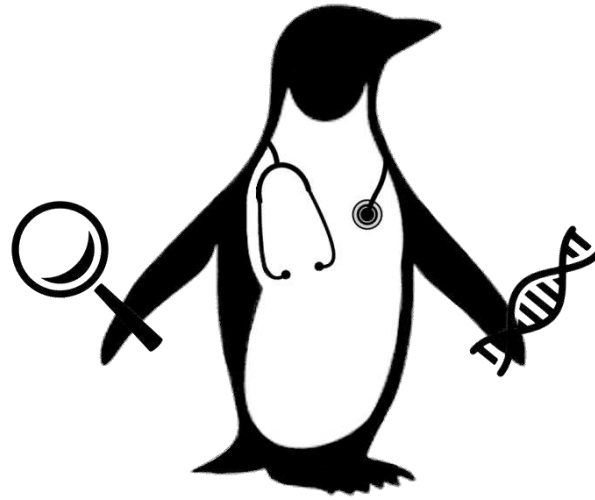


Ph.D. plan introduction

Impacts



Deliverables and goals



PENGQUIN

PErsoNal Genome QUery IN healthcare and clinical practice

Personal Genome Sequence Data

Human genome data is big...

~3.2 BILLION base pairs ----> ~1 GB

Sequencing output file ----> ~200 GB

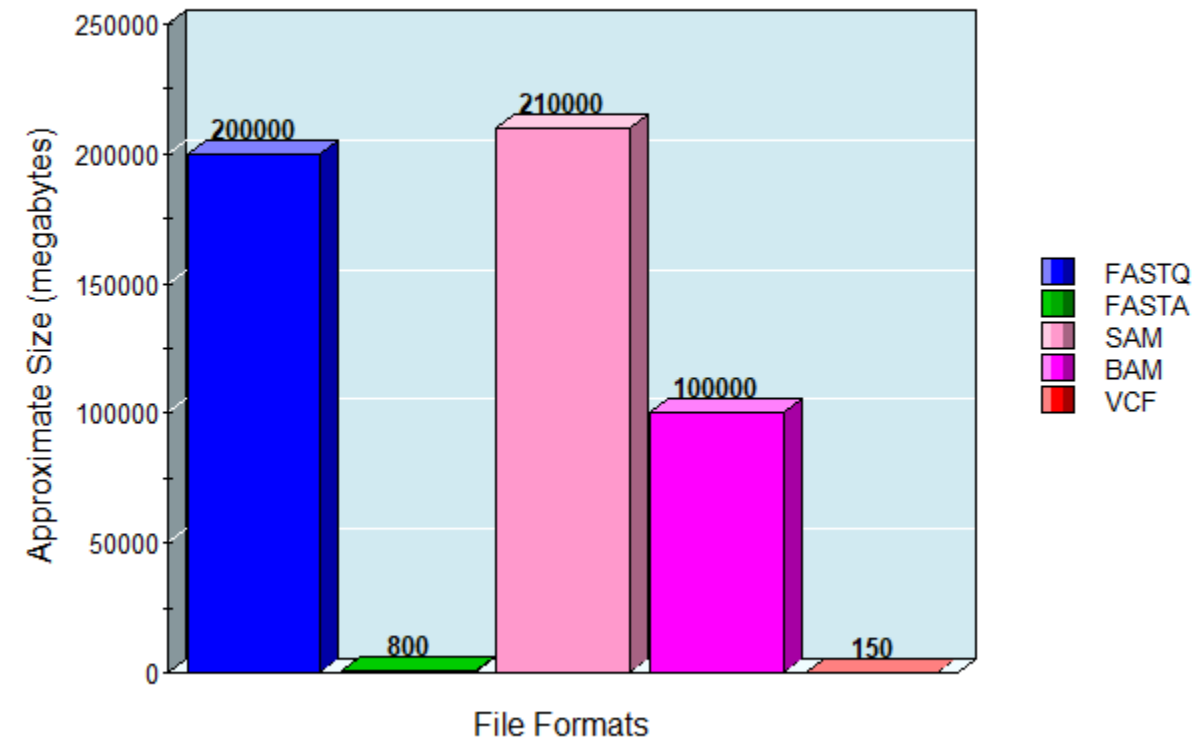
Variant Call Format (VCF) file ----> ~200 MB

- Individual humans exhibit ~3 million mutations
- VCF file **records only those differences**
- Flat file

Other considerations

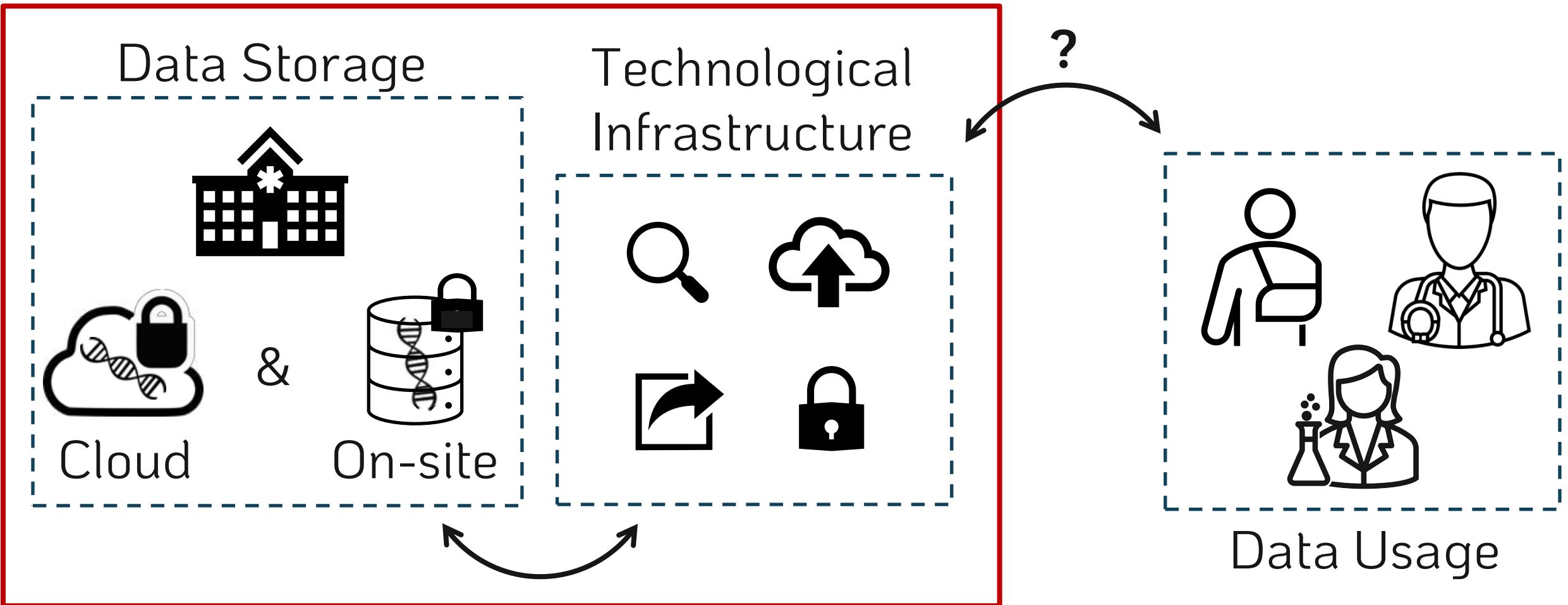
-  Storage
-  Privacy
-  Governance
-  Formatting

Sequencing File Sizes



Semantic Genome Data?

Institution-centric



Scaling Genome Data Usage

Research genomic data sharing initiatives:



State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US)
EU Health Data Spaces  Not much...

Solid Technology



- Does not solve ALL problems, but offers an environment where improvements are possible
 - ✓ Easier sharing (web-facilitated)
 - ✓ Granular privacy controls
 - ✓ Direct patient data access
 - ✓ Data representation flexibility (RDF)
 - ✓ Possibility of data-linking and querying

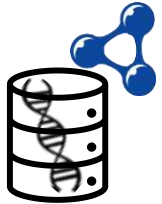
Research Questions



1. Can Solid be implemented for genomic data storage?



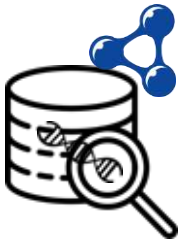
What is the impact on data policy enforcement?



2. Can (VCF) genomic data be represented as RDF?



Can genomic and other medical data be linked semantically?



3. Can RDF genomic data be queried in a useful way?



Can this improve genomic data discoverability for researchers?

Project Goals

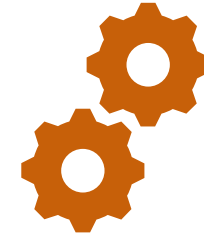


Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

While **exploring representation + querying limits**

Project Goals

But wait...



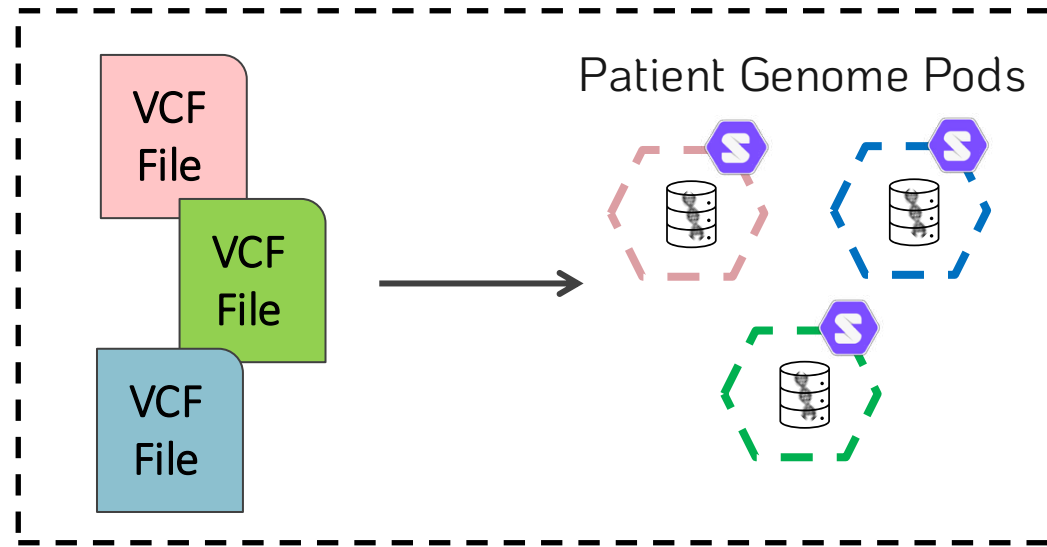
This sounds like an **ENGINEERING** project!

Where is the **RESEARCH**?



Workflow Overview

⚙️ Data Storage



Data Storage



Solid can be implemented for genomic data storage.



Solid pod instances → Community Solid Server¹

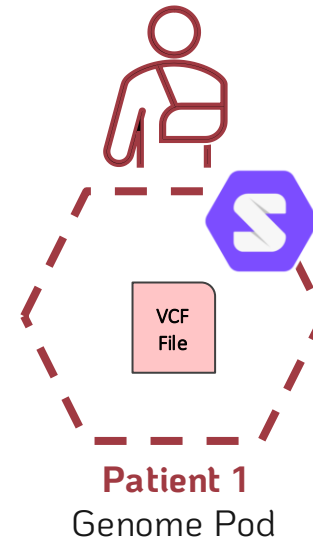


VCF Genomic Data → Illumina Platinum Genomes²

Patient 1
Genome Data

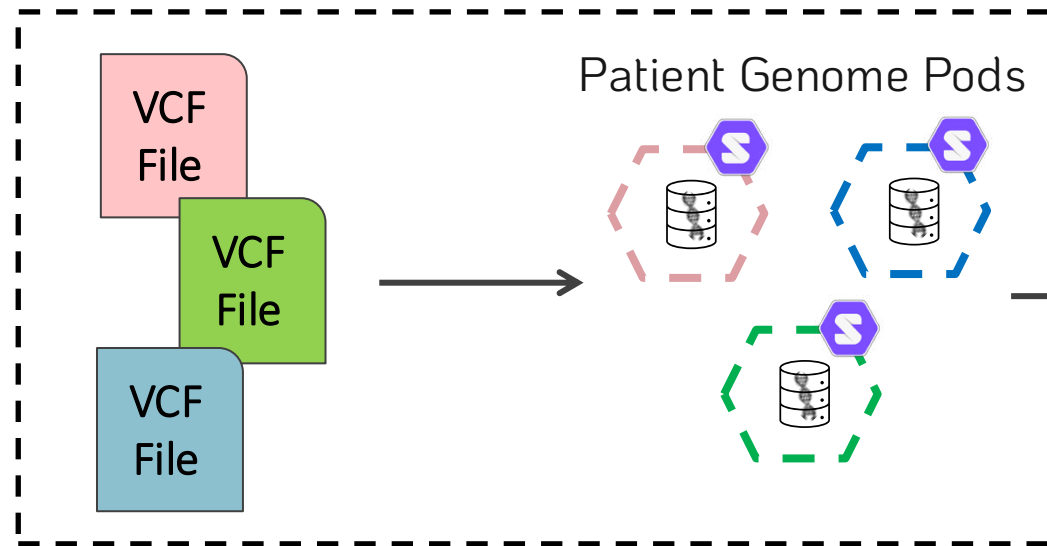


Data Ingestion

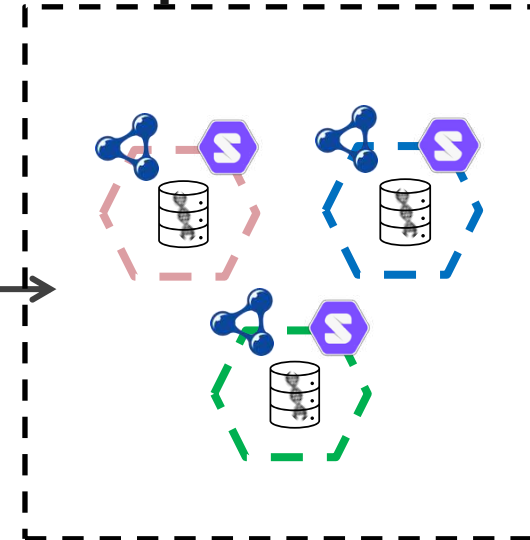


Workflow Overview

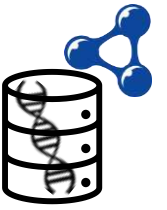
Data Storage



Representation



Genome Data in RDF

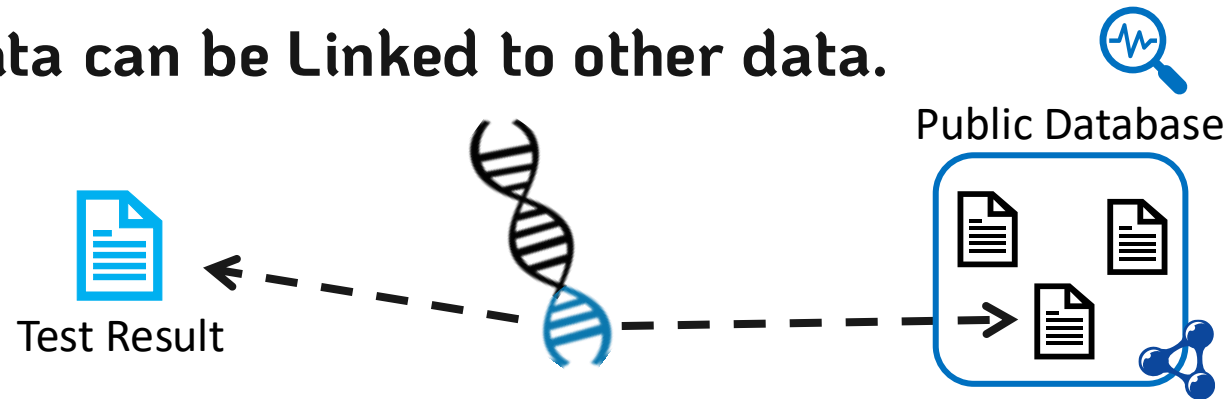


VCF genomic data can be represented as RDF.



 VCF to RDF Vocabularies exist $\longrightarrow$ SPHN Semantic Interoperability Framework³
+ FHIR HL7 Format⁴

VCF genomic data can be Linked to other data.



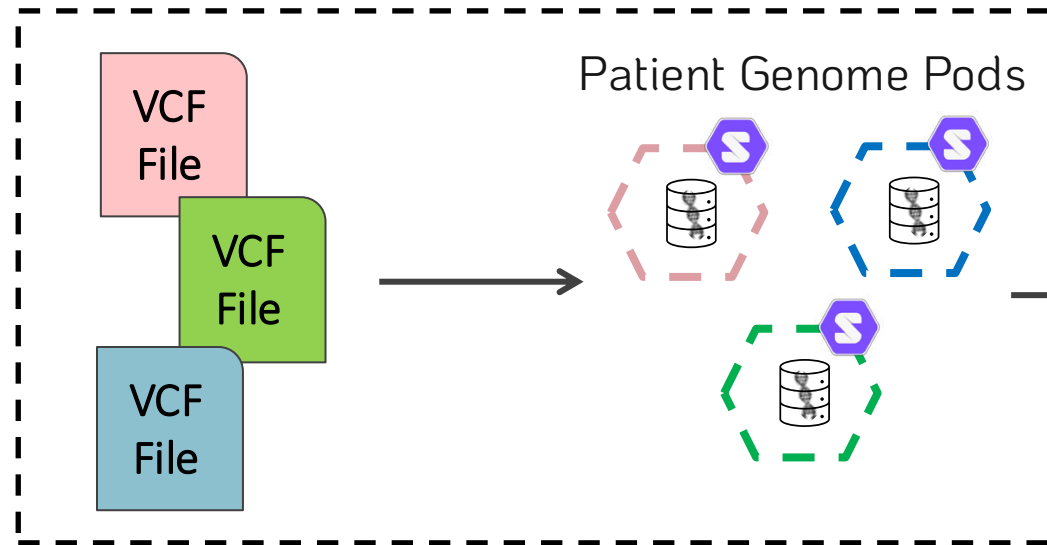
How to automate this?

What vocabularies could be used for the linkages?

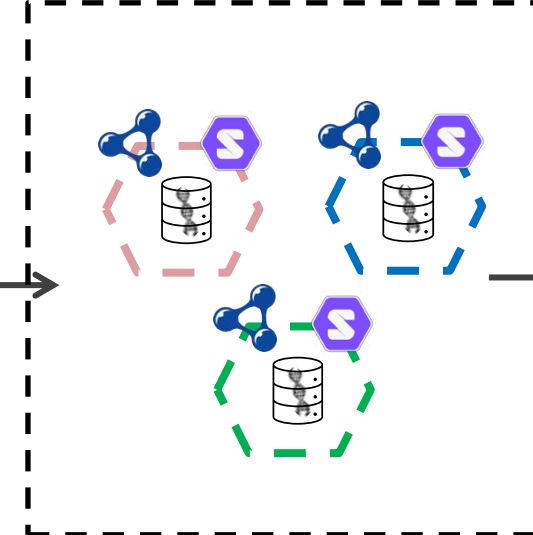


Workflow Overview

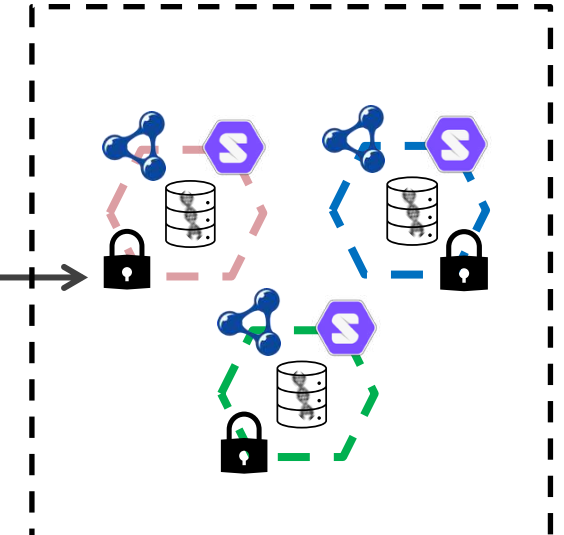
Data Storage



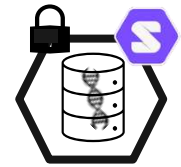
Representation



Privacy



Data Privacy Controls



Granular privacy controls can be implemented in Solid for genomic data.



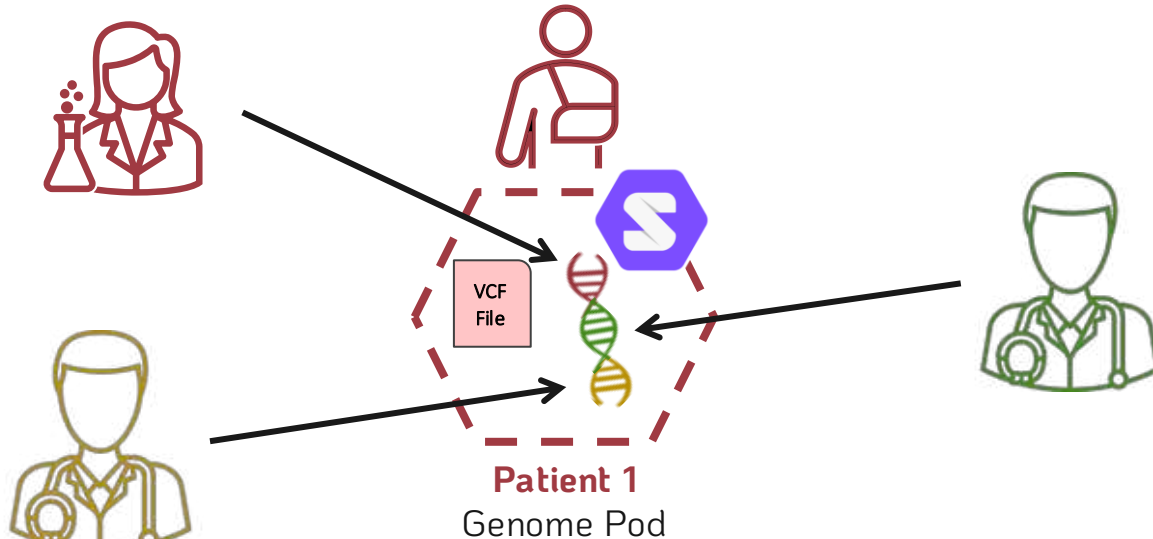
Web Access Control



Solid Protocol ACP⁵

How granular can we make the privacy controls?

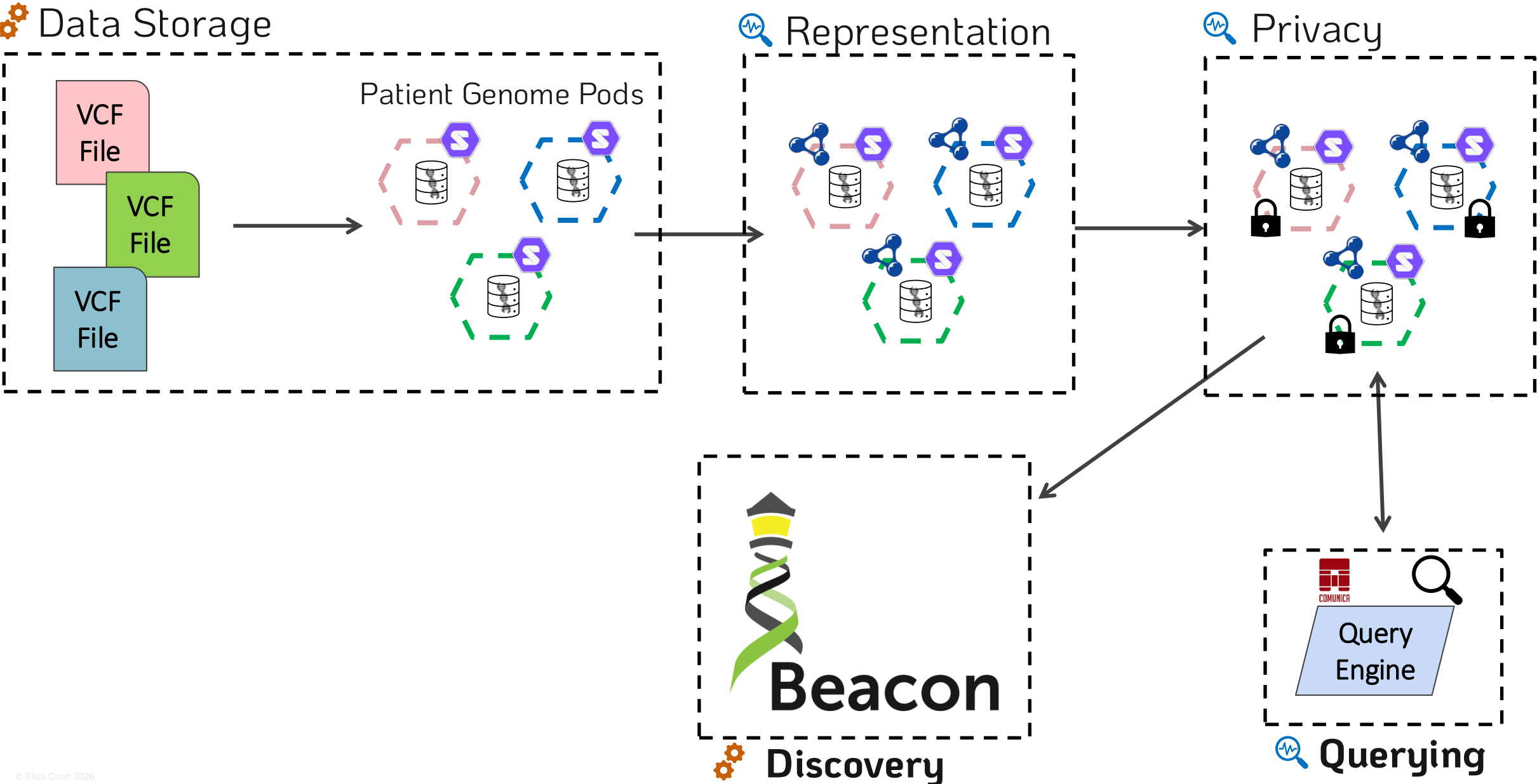
Privacy controls at chromosome level, or even mutation level.



Chromosome is a semantic Class

Mutation is also a semantic Class,
above a triple but below a Chromosome

Workflow Overview



Querying Genome Pods



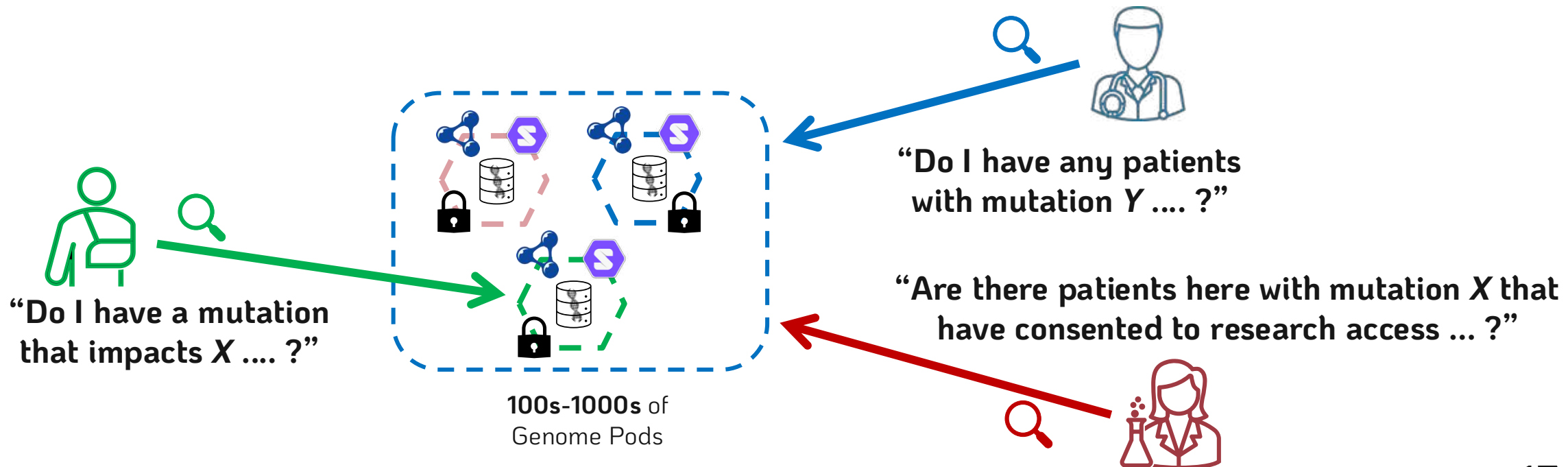
RDF genomic data pods can be queried.



Query Engine



Custom Comunica implementation/engine⁶



⁶ <https://comunica.dev/>

PhD Goals

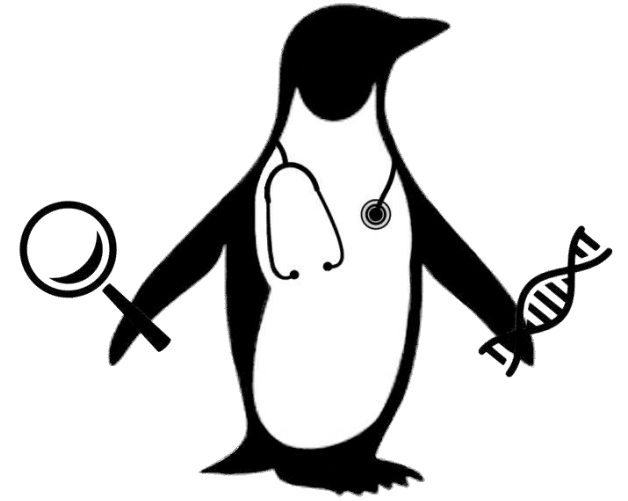
- ✓ Finished
- ⌚ Soon
- 🕒 Future

Basic Functionalities:

-  Genomic data storage in Solid pods ✓
-  Data privacy management ⌚
-  Data Sharing via the web ✓

Experimental research:

-  Represent genomic data as RDF? ⌚
-  Linking genomic and health data? 🕒
-  Single and Many pod Querying? 🕒



Questions?

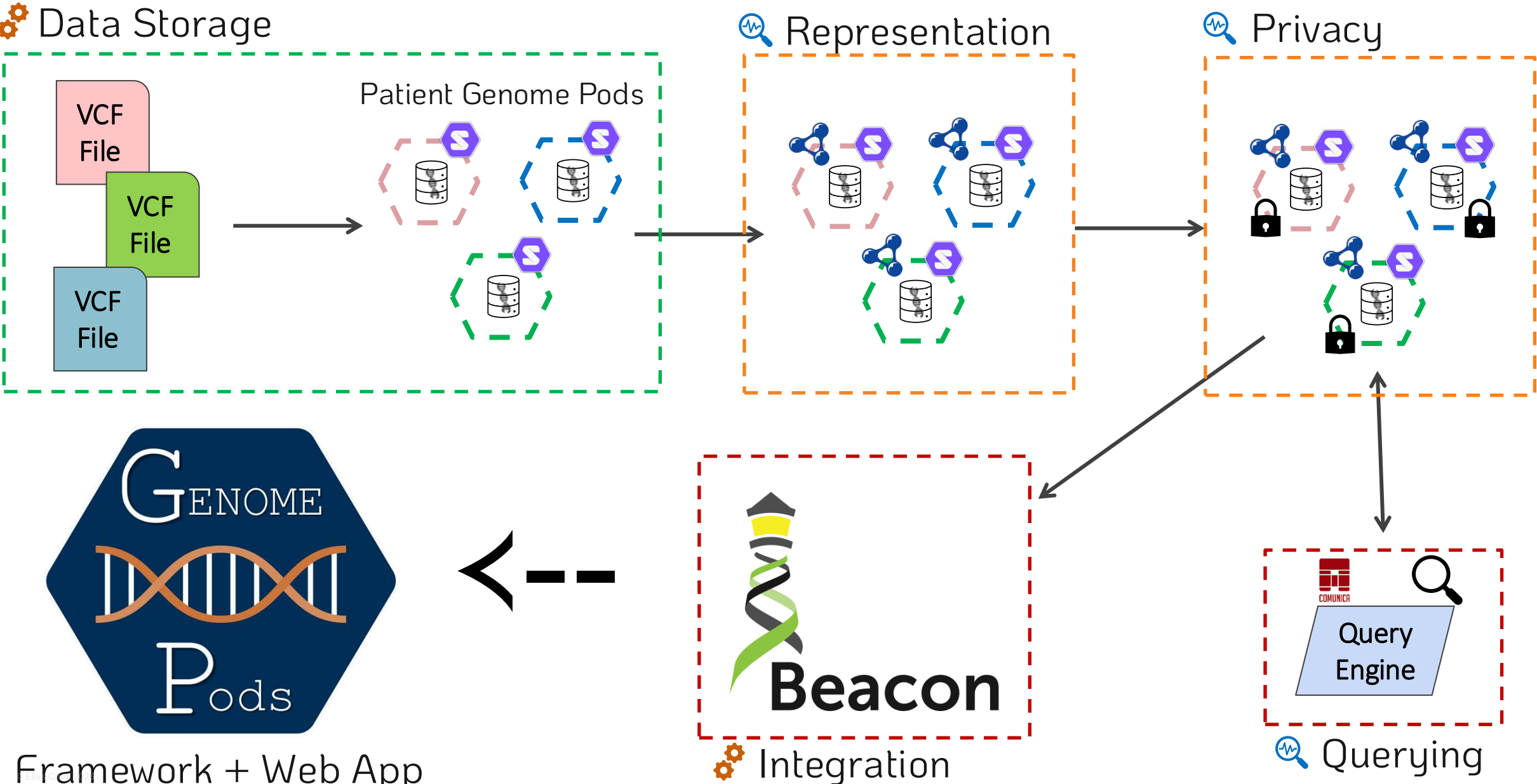
Paper



Contact



Workflow Overview



Querying Genome Pods (possibilities)



Single Pod Query

Utilizes an intra-pod index

Many Pod Query

First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed

